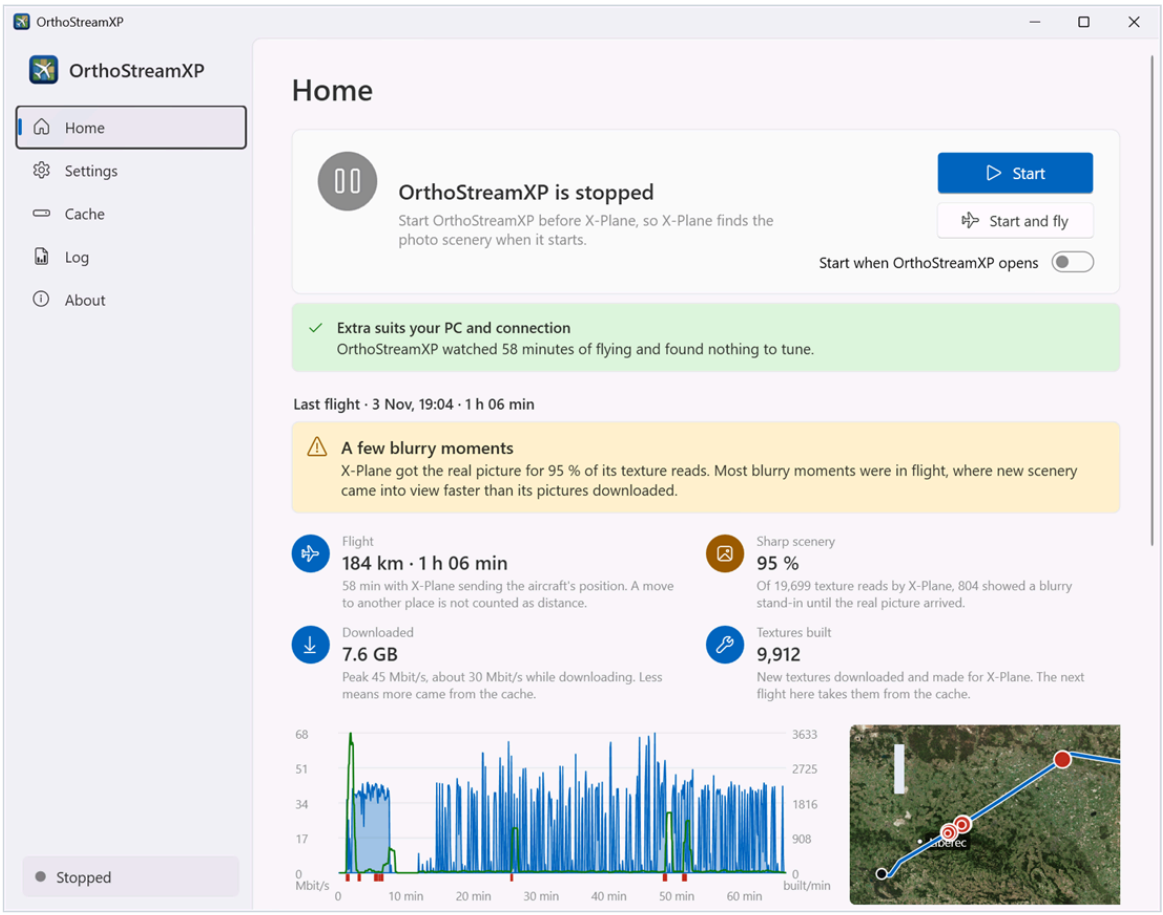
# OrthoStreamXP user guide

Version 1.0.0

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OrthoStreamXP adds real-world aerial photos to your X-Plane 12 scenery while you fly. It downloads the pictures for the area around your aircraft, turns them into X-Plane textures, and keeps them on your disk for your next flight.




The OrthoStreamXP window after a flight

## The short version

1. Install WinFsp, then OrthoStreamXP, then restart your computer.
2. Open OrthoStreamXP and finish the setup: X-Plane folder, scenery packages folder, cache folder, quality.
3. Every time you fly, click **Start and fly** in OrthoStreamXP. It starts the scenery first and then X-Plane.
4. Load a flight. A new area takes a few minutes the first time; after that it comes from the cache.

> **Remember:** start OrthoStreamXP **before** X-Plane. X-Plane looks for scenery only when it starts.

# Installation

## What you need

- **Windows 10 or 11 (64-bit).**
- **X-Plane 12.**
- **Scenery packages of your own** for the regions you fly. OrthoStreamXP adds the photos to them, but it does not make the terrain (the mesh) itself. Packages made by Ortho4XP, OrthoStudio XP, AutoOrtho or XPME work. Each package is a folder that contains an `Earth nav data` folder. Put all of them in **one parent folder**, for example:

  ```
  D:\OrthoPackages\
      Europe\
          Earth nav data\
          terrain\
          ...
      Asia\
          Earth nav data\
          ...
  ```

- **WinFsp**, a free Windows driver that OrthoStreamXP uses to show the scenery to X-Plane.
- **An internet connection** for areas you have not flown yet.
- **Disk space for the cache.** Each new 1° × 1° area you fly can take up to about 2 GB at Balanced, 8 GB at Sharp and 30 GB at Ultra or Extra. 25 GB is a reasonable minimum; a fast SSD is best.
- **A graphics card** with enough memory for the quality you choose (see Settings › Quality).

## 1. Install WinFsp

Download the latest WinFsp installer from winfsp.dev/rel and run it with the default options.

If you skip this step, the first setup shows a **Get WinFsp** button. Install WinFsp, restart your computer, then open OrthoStreamXP again.

## 2. Install OrthoStreamXP

1. Download `OrthoStreamXP-Setup.msi` from the Releases page.
2. Run it. The installer also installs the .NET 10 desktop runtime if your PC does not have it yet.

### Windows SmartScreen warning

The installer is not code-signed yet, so Windows may say it does not recognise the publisher. If you downloaded the file from the official Releases page, click **More info**, then **Run anyway**.

## 3. Restart your computer

Restart Windows after installing WinFsp and OrthoStreamXP. WinFsp's driver is ready only after a restart: until then OrthoStreamXP can fail to start the scenery, even though WinFsp shows as installed.

## 4. Open OrthoStreamXP

Start OrthoStreamXP from the Start menu. The first time, it opens the first setup.

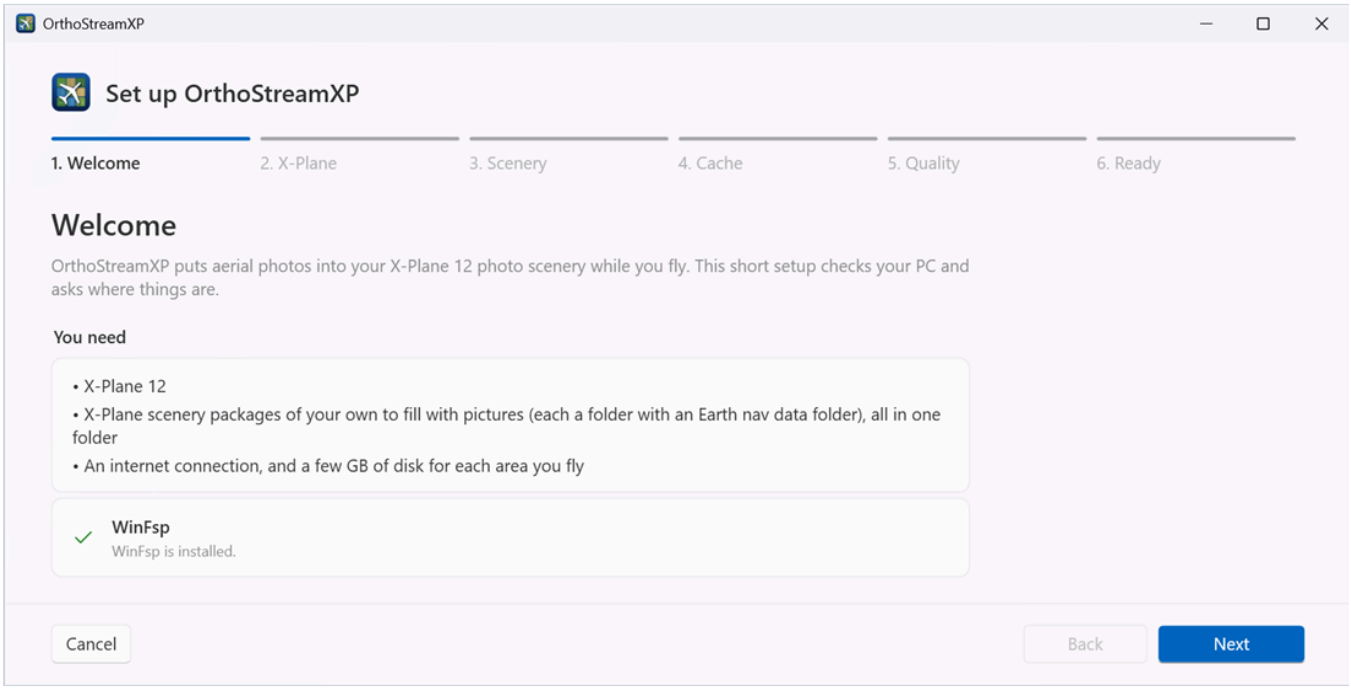
Next: First setup

# First setup

The setup opens the first time you start OrthoStreamXP. It has six steps and takes a minute or two. Use **Next** and **Back** to move between the steps; nothing is saved until you click **Finish**.

You can run it again later from **Settings › Run the setup again**. A second run starts from your current settings, and **Cancel** leaves them unchanged.

## 1. Welcome

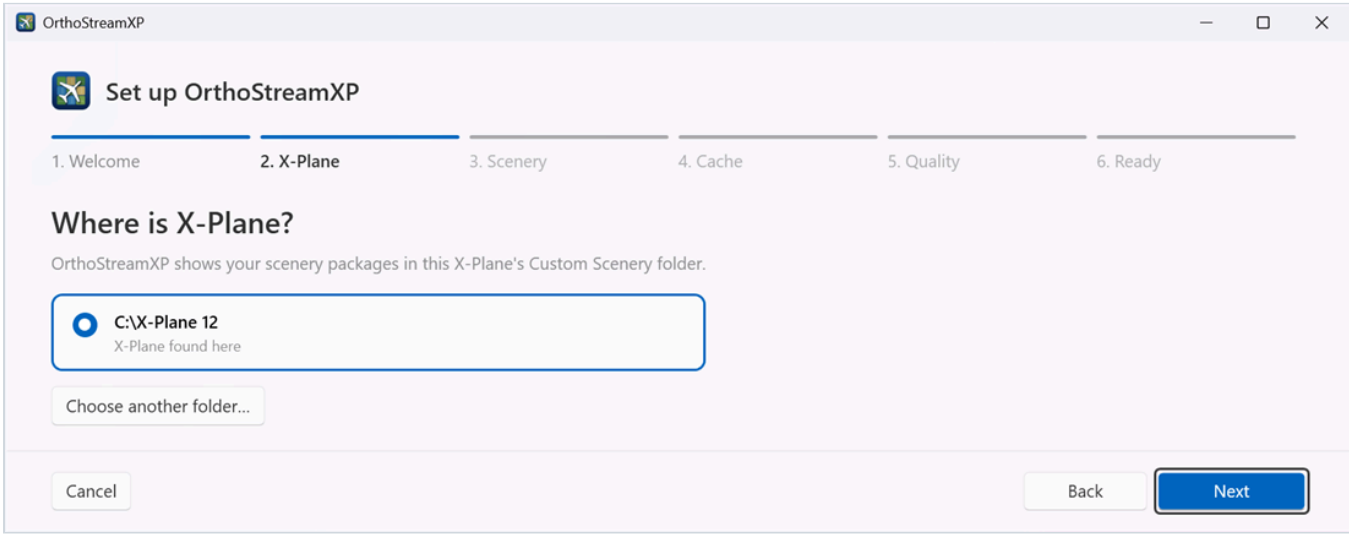



Setup step 1: Welcome

This step lists what you need and checks for WinFsp.

- A green tick with **WinFsp is installed** means you can continue.
- If WinFsp is missing, click **Get WinFsp** and install it. Then restart your computer and open OrthoStreamXP again: WinFsp works only after a restart.

## 2. X-Plane

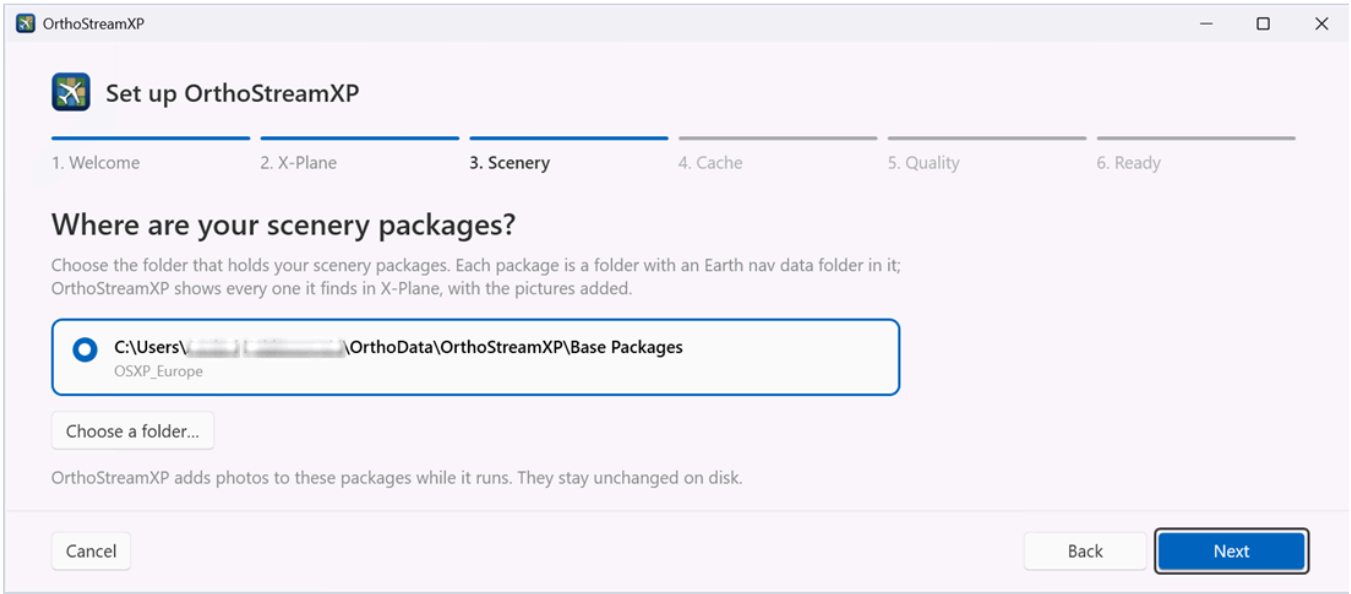



Setup step 2: X-Plane

OrthoStreamXP looks for X-Plane 12 on your PC and offers every installation it finds. Pick the one you fly with.

If yours is not listed, click **Choose another folder...** and choose the folder that contains `X-Plane.exe`. That folder must also have a `Custom Scenery` folder.

## 3. Scenery




Setup step 3: Scenery packages

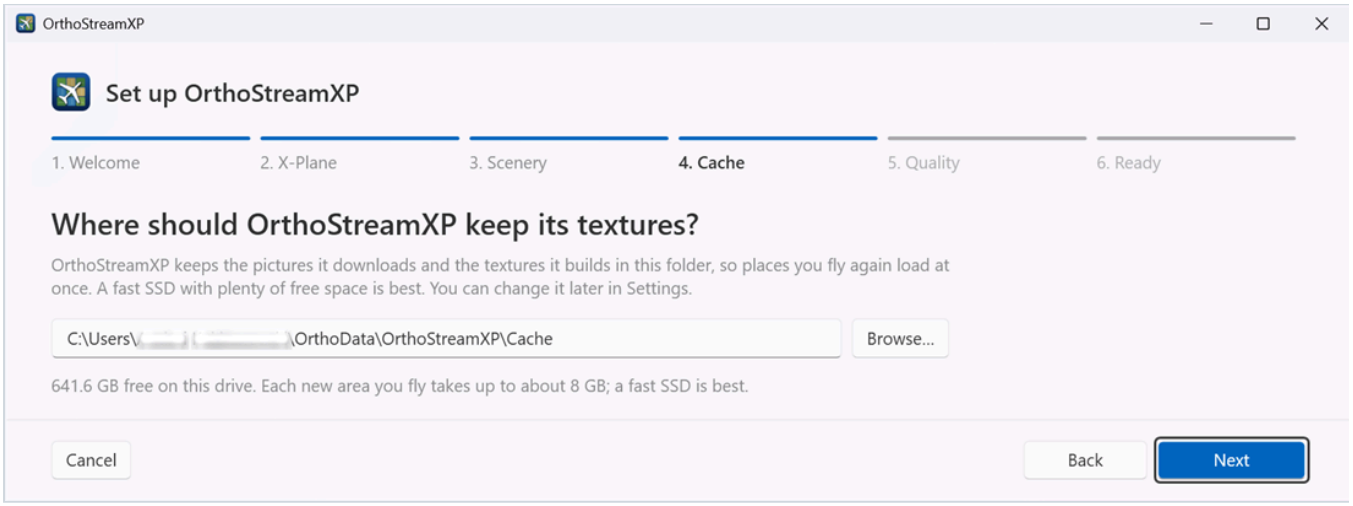
Click **Choose a folder...** and pick the **parent folder** that holds your scenery packages, not one of the packages. Below the folder, OrthoStreamXP lists the packages it found.

In the example below you have to choose the “OrthoPackages” folder.

```
D:\OrthoPackages\
    Europe\
        Earth nav data\
        terrain\
        ...
    Asia\
        Earth nav data\
        ...
```

- Your packages stay unchanged on disk. While OrthoStreamXP runs, they appear in X-Plane's `Custom Scenery` folder with the photos added. When it stops, they disappear from there again.
- If the folder has no packages, the hint turns red: *That folder has no scenery packages*. Check that each package has an `Earth nav data` folder inside it.
- If X-Plane's `Custom Scenery` already contains a folder with the same name as one of your packages, you get a warning. Usually another streaming tool still shows that package there; close that tool before you start OrthoStreamXP.

## 4. Cache

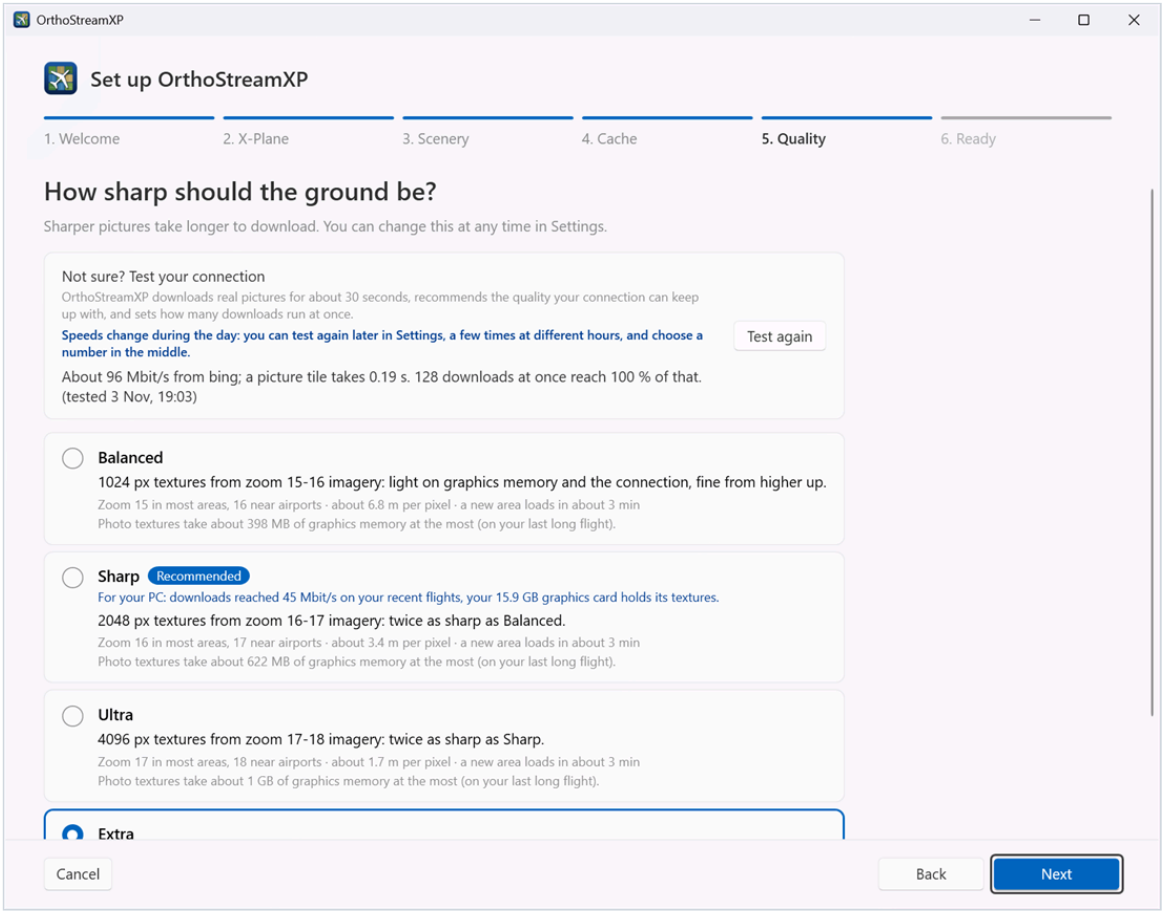



Setup step 4: Cache folder

This is where OrthoStreamXP keeps the pictures it downloads and the textures it builds, so areas you fly again load at once.

- Click **Browse...** or type a full path, such as `D:\OrthoStreamXP\Cache`.
- Choose a folder on a fast SSD with plenty of free space. The page shows how much space is free on that drive. Each new 1° × 1° area takes up to about 2 GB at Balanced, 8 GB at Sharp and 30 GB at Ultra or Extra.
- You can change the folder or set a size limit later on the Cache page.

## 5. Quality




Setup step 5: Quality

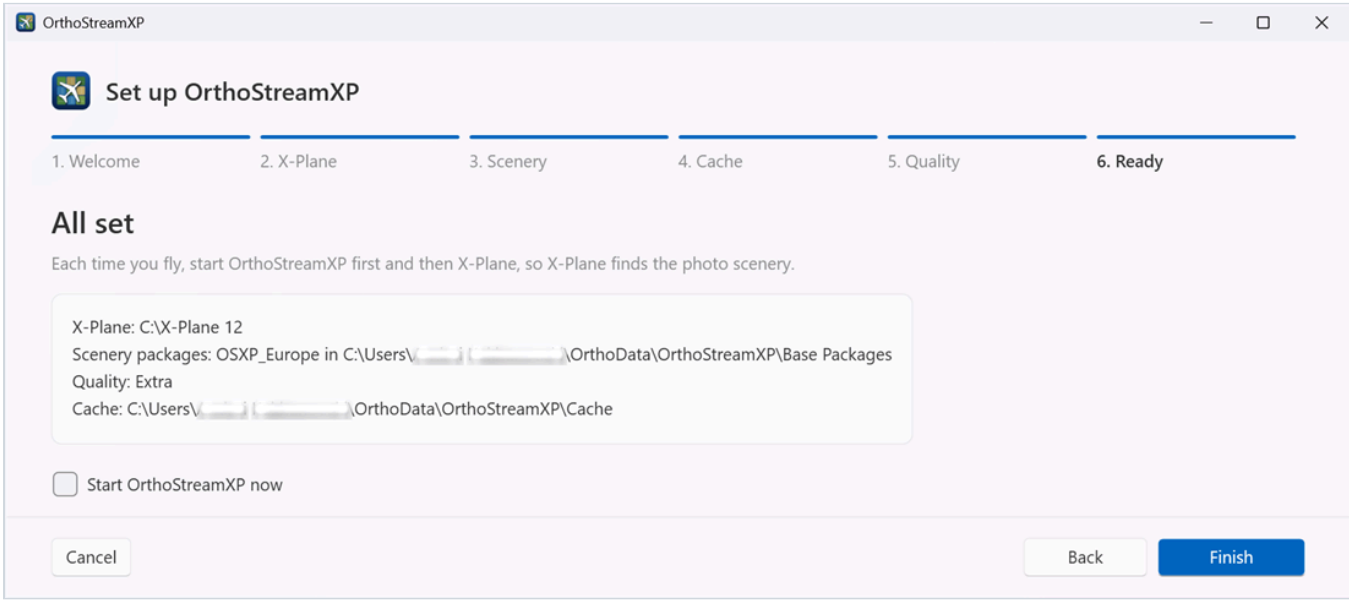
Pick how sharp the ground should be: **Balanced**, **Sharp**, **Ultra** or **Extra**. Each card shows the texture size, the imagery zoom level, the metres per pixel, and roughly how long a new area takes to load. See Settings › Quality for a comparison.

**Not sure?** Click **Test my connection** (it says **Test again** if you have tested before). OrthoStreamXP downloads real pictures for about 30 seconds, then:

- marks the quality your connection can keep up with as **Recommended**, and selects it for you;
- sets how many downloads run at once.

Internet speed changes during the day, so you can repeat the test later in Settings › Pictures and download.

## 6. Ready




Setup step 6: Ready

This step summarises your choices: the X-Plane folder, the scenery packages, the quality and the cache folder. Tick **Start OrthoStreamXP now** if you want to fly straight away, then click **Finish**.

Next: Flying

# Flying

## The start order

X-Plane looks for scenery only when it starts, so **OrthoStreamXP must be running before X-Plane starts**.

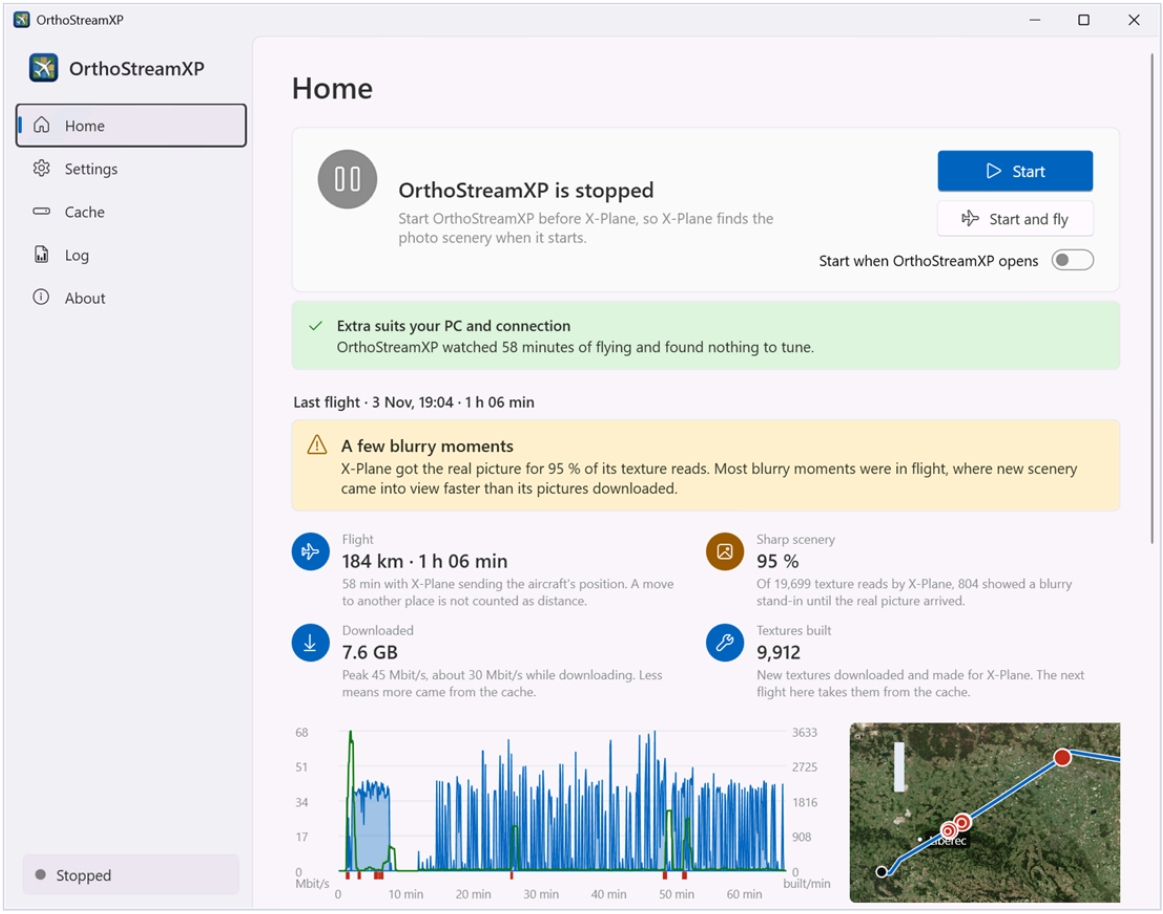
The Home page has two ways to start:

| Button | What it does |
|---|---|
| **Start and fly** | Starts OrthoStreamXP, waits until the scenery is ready, then starts X-Plane for you. This is the easiest option. While OrthoStreamXP runs, the button reads **Start X-Plane**. |
| **Start** | Starts only OrthoStreamXP. Start X-Plane yourself afterwards. |

Turn on **Start when OrthoStreamXP opens** if you want the scenery to start every time you open the app.

If X-Plane was already running when you started OrthoStreamXP, Home shows **Restart X-Plane**. Quit X-Plane and start it again to see the photos.

## The Home page




The Home page

Before your first start, Home shows **How it works**: start OrthoStreamXP, start X-Plane, fly.

From top to bottom:

- **Status card.** Shows whether OrthoStreamXP is stopped, starting, ready (*Photo scenery is ready*) or in use (*Flying with photo scenery*). The **Start** button becomes **Stop** while it runs. The dot at the bottom of the sidebar shows the same state on every page.
- **Notices.** Tuning suggestions, updates and problems appear here, each with a button for the next step.
- **Last flight.** A report on your most recent flight, shown while OrthoStreamXP is stopped (see below).
- **Session figures** (*This session* while it runs, *Last session* after it stops): how many textures X-Plane got, how many of its reads got the real picture, how many textures are being built now, and the download speed or the number of picture tiles downloaded.
- **X-Plane.** *Waiting for X-Plane* until X-Plane sends the aircraft position, then *Connected* with the aircraft's position and speed. *Paused or in the menu* means X-Plane has stopped sending it for now. There is nothing to set up in X-Plane: OrthoStreamXP finds it on the same PC by itself.
- **Quality.** Your current preset, with a **Change quality** link.
- **Scenery packages.** The packages shown in X-Plane while OrthoStreamXP runs.

## During the flight

- **Load your flight as usual.** OrthoStreamXP builds the textures X-Plane asks for, and follows your aircraft to build the scenery ahead of it.

- **A new area takes a few minutes the first time.** Until its pictures arrive, X-Plane briefly shows a blurrier version of the ground. The next time you fly there, the area comes straight from the cache.
- **Sharper presets take longer.** The quality cards in Settings show roughly how long a new area takes to load.
- **Stopping.** When you finish flying, quit X-Plane first, then click **Stop** in OrthoStreamXP. OrthoStreamXP finishes the textures it is building and removes the photo scenery from X-Plane. If you close the window while OrthoStreamXP runs, it asks first, because X-Plane would lose the photo scenery.

## After the flight: the flight report

When OrthoStreamXP stops, Home shows a report of the flight. It starts with a verdict:

- **Smooth flight:** the scenery was sharp, or nearly so.
- **A few blurry moments** or **Often blurry**, with when they happened. Blurry moments at the start come from X-Plane loading the area and clear as soon as the pictures arrive. Blurry moments in flight mean new scenery came into view faster than its pictures downloaded; if that happens often in one place, a lower quality or the tuning suggestions may help.

Then the figures:

- **Flight:** distance and duration. Only the time X-Plane sent the aircraft's position counts, and moving the aircraft to another place is not counted as distance.
- **Sharp scenery:** the share of X-Plane's texture reads that got the real picture straight away. Lower values mean more blurry moments.
- **Downloaded:** data downloaded and the peak speed. Less data means more came from the cache.
- **Textures built:** new textures made for X-Plane during this flight.
- **A chart** of download speed and textures built over time, with blurry moments marked in red.
- **A map** of your route, with the places where the scenery was blurry.

## Tuning suggestions

After a flight of **10 minutes or more**, OrthoStreamXP checks how the scenery kept up on your PC and connection.

- If your preset suits your setup, Home says so, for example *Extra suits your PC and connection*.
- If changes would help, Home shows **Tuning for from your last flight**. It lists each suggested change and why it would help. Click **Apply** to accept the changes or **No, thanks** to ignore them. Nothing changes without your OK.
- After a shorter flight, Home shows *Fly a little longer for tuning*.

You can undo the tuning at any time in Settings › Tuning.

Next: Settings

# Settings

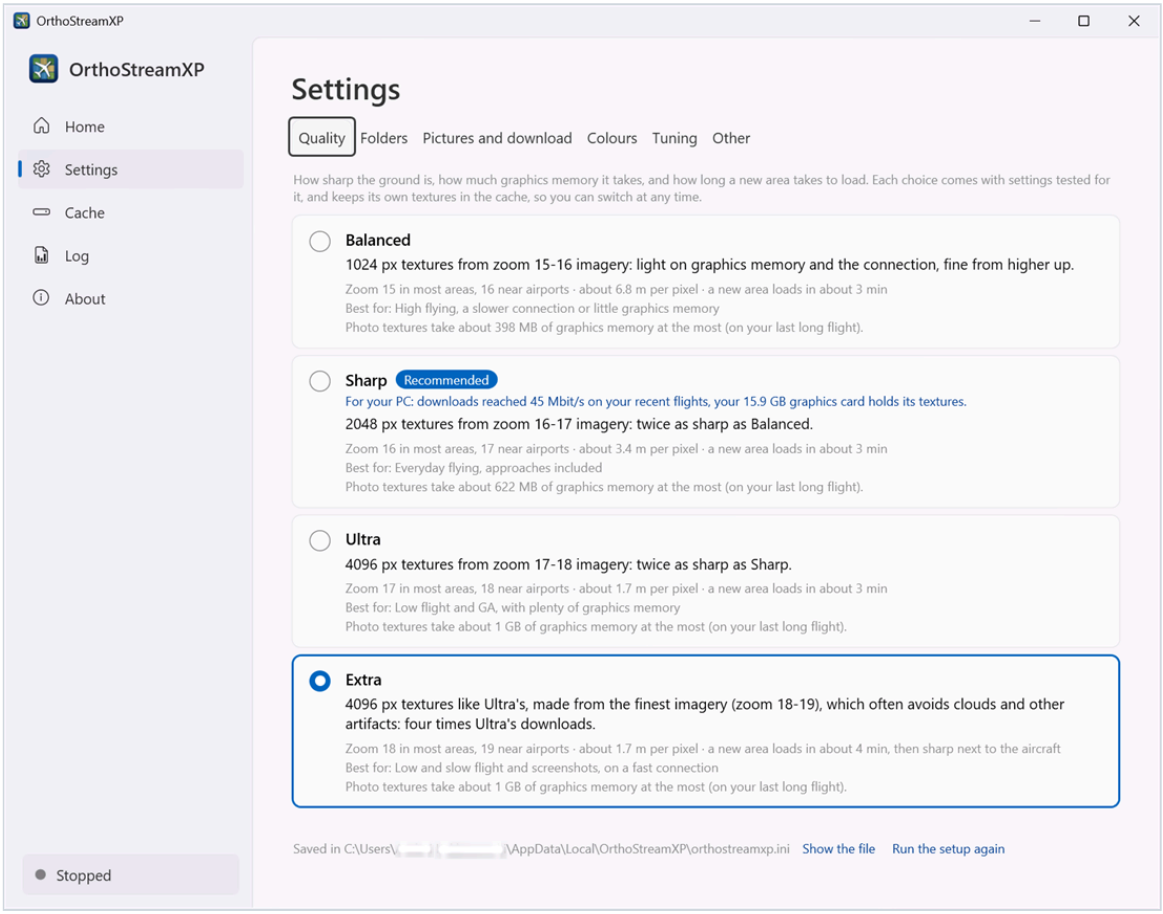
The Settings page has six tabs. Changes are saved straight away, and a short **Saved** note confirms each one.

Settings can be changed only while OrthoStreamXP is stopped. While the scenery is streaming, the page shows the settings in use but you cannot edit them: a banner says *Streaming is running. Stop it to change settings.* Click **Stop streaming** in the banner to stop it. Changes apply from the next start.

At the bottom of every tab:

- **Show the file** shows the settings file (`%LOCALAPPDATA%\OrthoStreamXP\orthostreamxp.ini`) in Explorer.
- **Run the setup again** reopens the first setup.

## Quality




Settings: Quality

Choose how sharp the ground is. Each preset comes with settings tested for it and keeps its own textures in the cache, so you can switch at any time without losing anything.

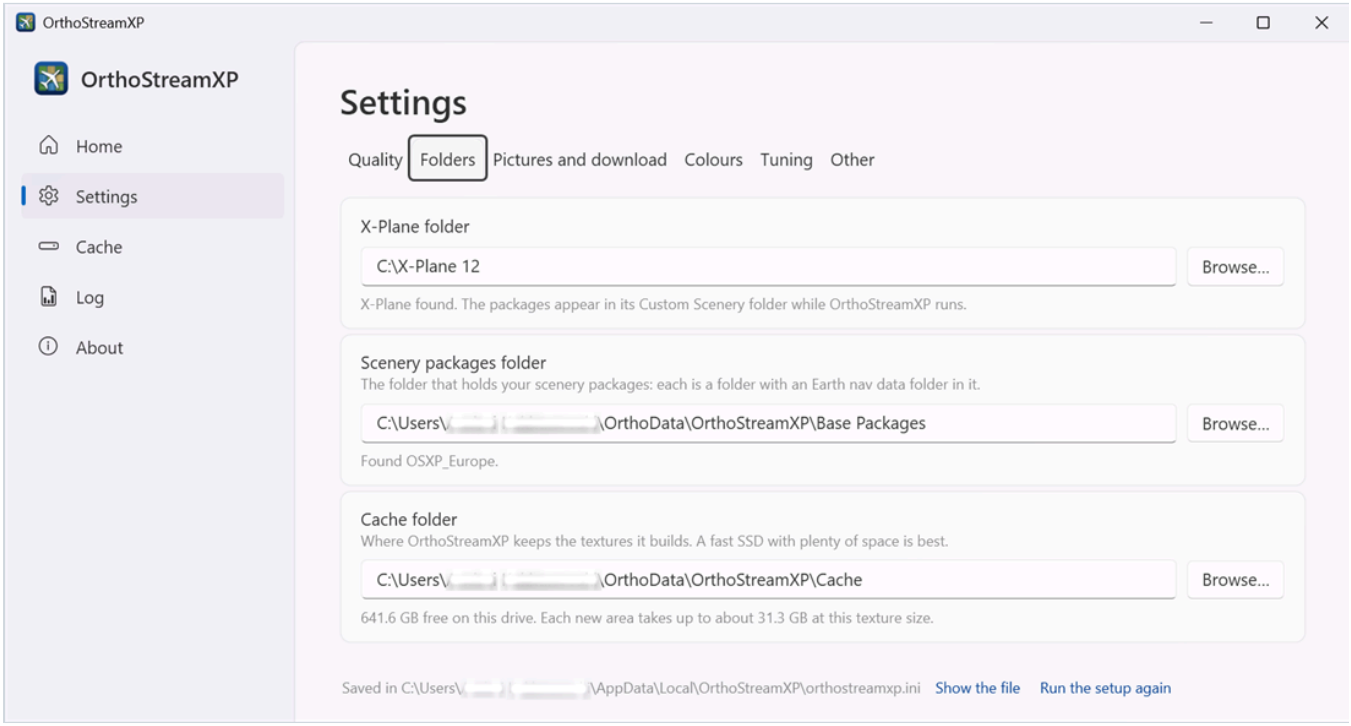
| Preset | Textures | Imagery zoom | Ground detail | Best for |
|---|---|---|---|---|
| **Balanced** | 1024 px | 15, 16 near airports | ~6.8 m per pixel | High flying, a slower connection or little graphics memory |
| **Sharp** | 2048 px | 16, 17 near airports | ~3.4 m per pixel | Everyday flying, approaches included |
| **Ultra** | 4096 px | 17, 18 near airports | ~1.7 m per pixel | Low flight and GA, with plenty of graphics memory |
| **Extra** | 4096 px | 17, 19 near airports | ~1.7 m per pixel | Low and slow flight and screenshots, on a fast connection |

**Extra** is Ultra with the finest imagery (zoom 19) near airports, which often avoids clouds and other artefacts. Elsewhere it is the same as Ultra. Near airports it downloads four times as much as Ultra, and a new area takes a little longer to load: about 4 minutes, then sharp next to the aircraft.

Each card shows the imagery zoom, the metres per pixel, roughly how long a new area takes to load, and how much graphics memory the photo textures take. Until you have flown, that memory figure comes from a test flight; after your first long flight, it is measured on your PC.

- The **Recommended** badge marks the preset that suits your connection and graphics card, with the reason below it.
- If a preset's photo textures plus X-Plane's own needs (about 5 GB) would be more than your graphics card has, its card shows a warning. X-Plane may then stutter or lower its texture detail; a smaller preset is safer.

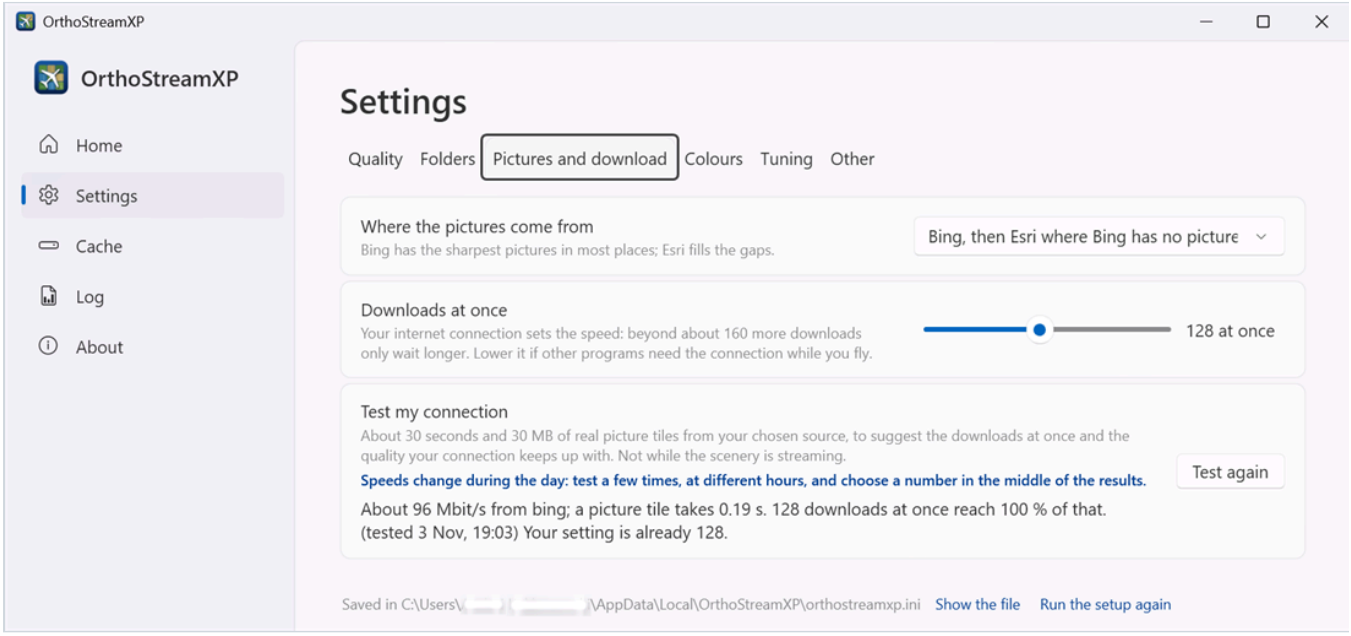
## Folders




Settings: Folders

- **X-Plane folder:** the X-Plane installation where the packages appear (in its `Custom Scenery` folder).
- **Scenery packages folder:** the parent folder of your packages. The line below it lists the packages found.
- **Cache folder:** where the textures are kept. The line below it shows the free space and how much a new area takes at the current texture size.

## Pictures and download

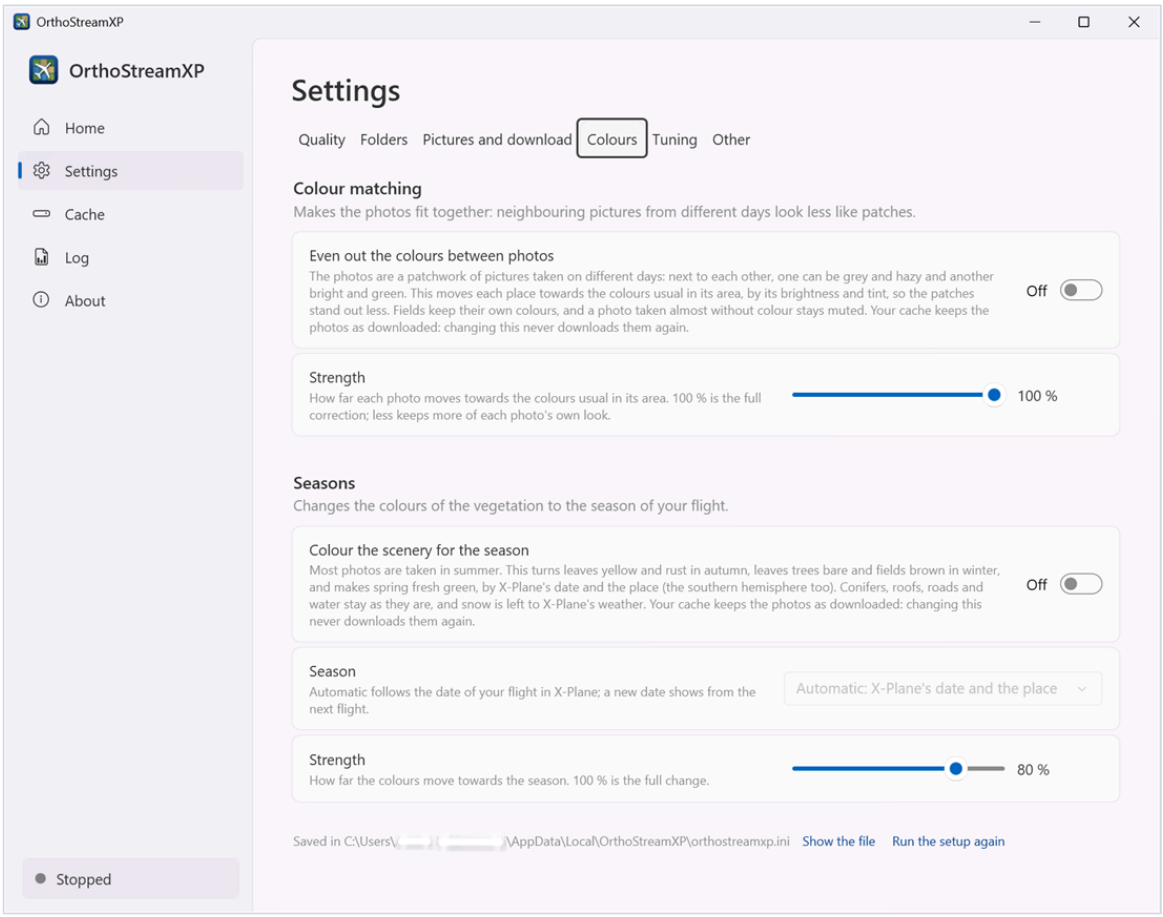



Settings: Pictures and download

- **Where the pictures come from:** the imagery source.

    - *Bing, then Esri where Bing has no picture* (recommended): the sharpest pictures in most places.
    - *Bing only*.
    - *Geoportal in Poland (national orthophoto), Bing and Esri elsewhere*.
- **Downloads at once:** how many pictures download in parallel, from 16 to 256. Your internet connection sets the real speed; beyond about 160, more downloads only wait longer. Lower this if other programs need the connection while you fly.
- **Test my connection:** about 30 seconds and 30 MB of real picture tiles from your chosen source. It suggests a value for *Downloads at once* and the quality your connection can keep up with. When its suggestion differs from your setting, a **Use N at once** button applies it. The test cannot run while the scenery is streaming. Speeds change during the day, so test a few times at different hours and choose a number in the middle of the results.

## Colours




Settings: Colours

### Preview

The top of the tab shows a photo from your cache twice: **Before** (as downloaded) and **After** (as X-Plane will get it with the settings below). It updates as you change them, and **Another photo** picks a different one. The preview needs a cache folder with at least one flown area in it. It works while the scenery is streaming too.

### Colour matching

Aerial photos are a patchwork of pictures taken on different days, so one patch can look grey and hazy next to a bright green one.

- **Even out the colours between photos** moves each place towards the colours usual in its area, so the patches stand out less. Fields keep their own colours.
- **Strength** sets how far the colours move: 100 % is the full correction, and less keeps more of each photo's own look.

### Seasons

Most aerial photos are taken in summer.

- **Colour the scenery for the season** changes the vegetation to match X-Plane's date and your location (southern hemisphere included):
  - **Spring:** fresh green.
  - **Autumn:** leaves turn yellow and rust.
  - **Winter:** deciduous trees look bare and fields turn brown.

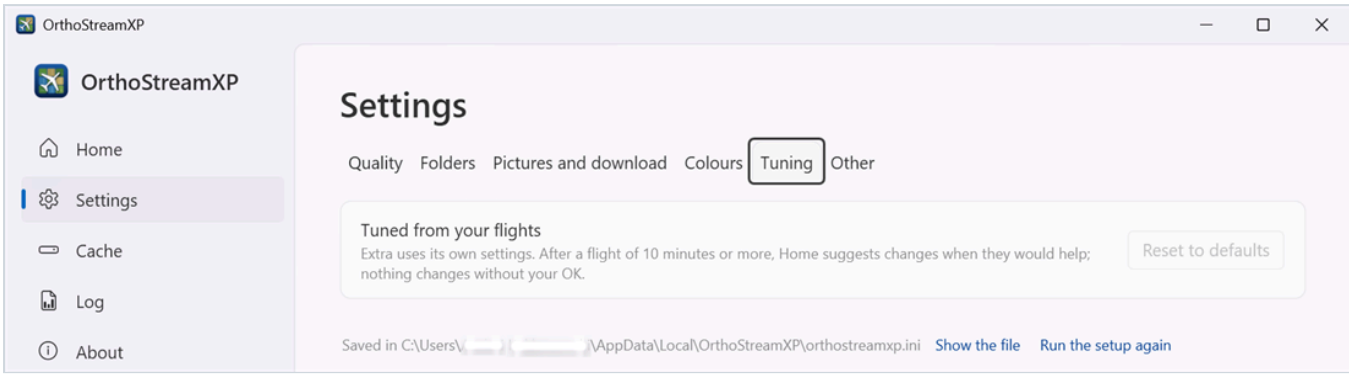
  Conifers, roofs, roads and water keep their colours, and snow is left to X-Plane's weather.
- **Season:** *Automatic* follows X-Plane's date and the place; a new date takes effect from the next flight. You can also force one season everywhere: *Early spring*, *Spring*, *Summer* (the pictures as they are), *Early autumn*, *Autumn*, *Late autumn* or *Winter, without snow*.
- **Strength:** how far the colours move towards the season. 100 % is the full change.

### Exposure

**Exposure** darkens photos that look too bright next to X-Plane's own scenery, like a camera's exposure. 100 % shows them as downloaded; 50 % is one stop darker.

All the colour options only change how the photos look. The cache keeps the photos as downloaded, so changing them never downloads anything again.

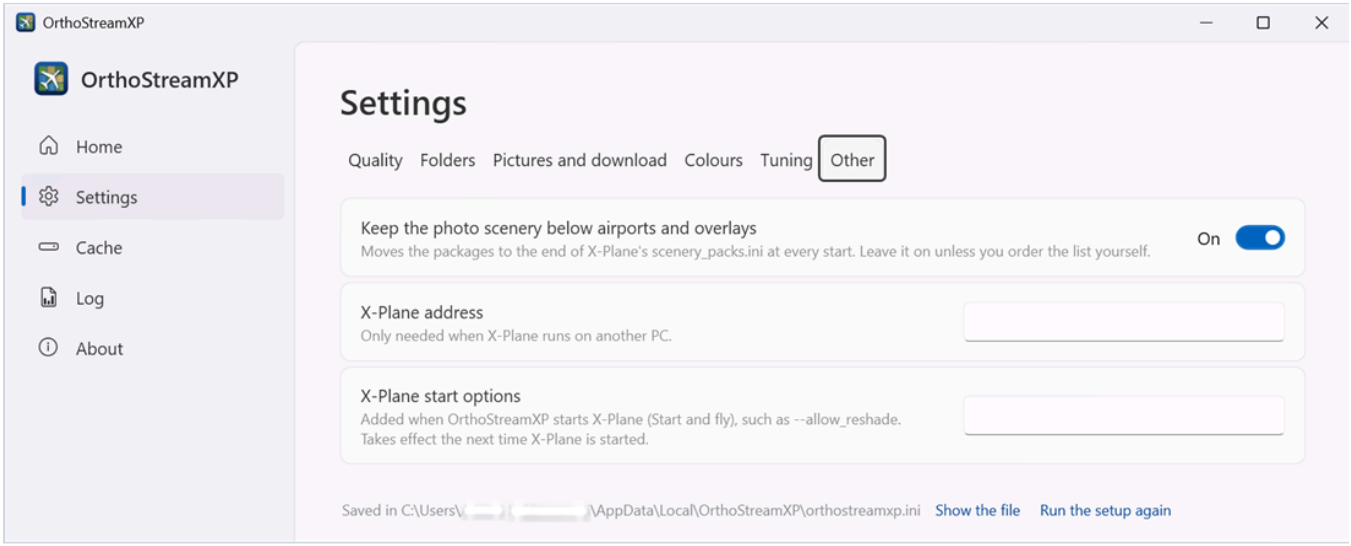
## Tuning




Settings: Tuning

After flights of 10 minutes or more, Home may suggest settings for your preset (see Flying › Tuning suggestions). The tab lists each setting the tuning changed, with the preset's own value beside it. **Reset to defaults** puts every setting back to the preset's tested values and forgets what was tuned; it asks first. The folders, the picture source and *Downloads at once* stay as they are.

## Other



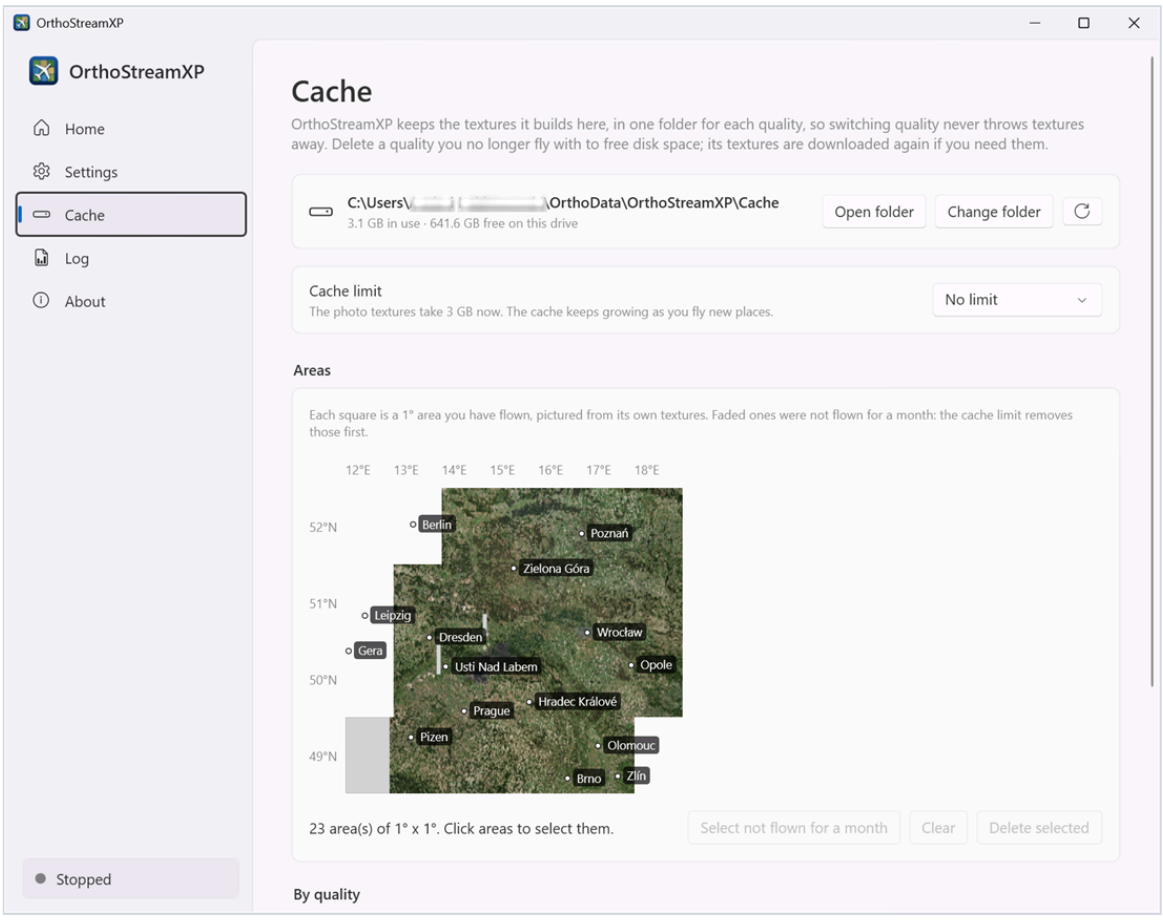

Settings: Other

- **Keep the photo scenery below airports and overlays:** at every start, moves the packages to the end of X-Plane's `scenery_packs.ini`, so airports and overlays draw on top of the photos. Leave it on unless you order that list yourself.
- **X-Plane address:** needed only when X-Plane runs on another PC.
- **X-Plane start options:** command-line options added when **Start and fly** starts X-Plane, for example `--allow_reshade`. They take effect the next time X-Plane starts.

Next: Cache

# Cache

The cache holds the pictures OrthoStreamXP downloads and the textures it builds. Areas you have flown before load from the cache at once, without the internet. Anything you delete is downloaded again the next time you fly there.




The Cache page

## Folder and limit

- The top card shows the cache folder, how much it uses and the free space on that drive.
  - **Open folder** opens the cache folder in Explorer.
  - **Change folder** chooses another cache folder.
  - The round arrow refreshes the figures.
- **Cache limit** caps the space the textures may take: *No limit*, 25 GB, 50 GB, 100 GB, 200 GB, 500 GB or 1 TB. When the cache grows past the limit, OrthoStreamXP deletes the areas you flew longest ago first.

## Areas

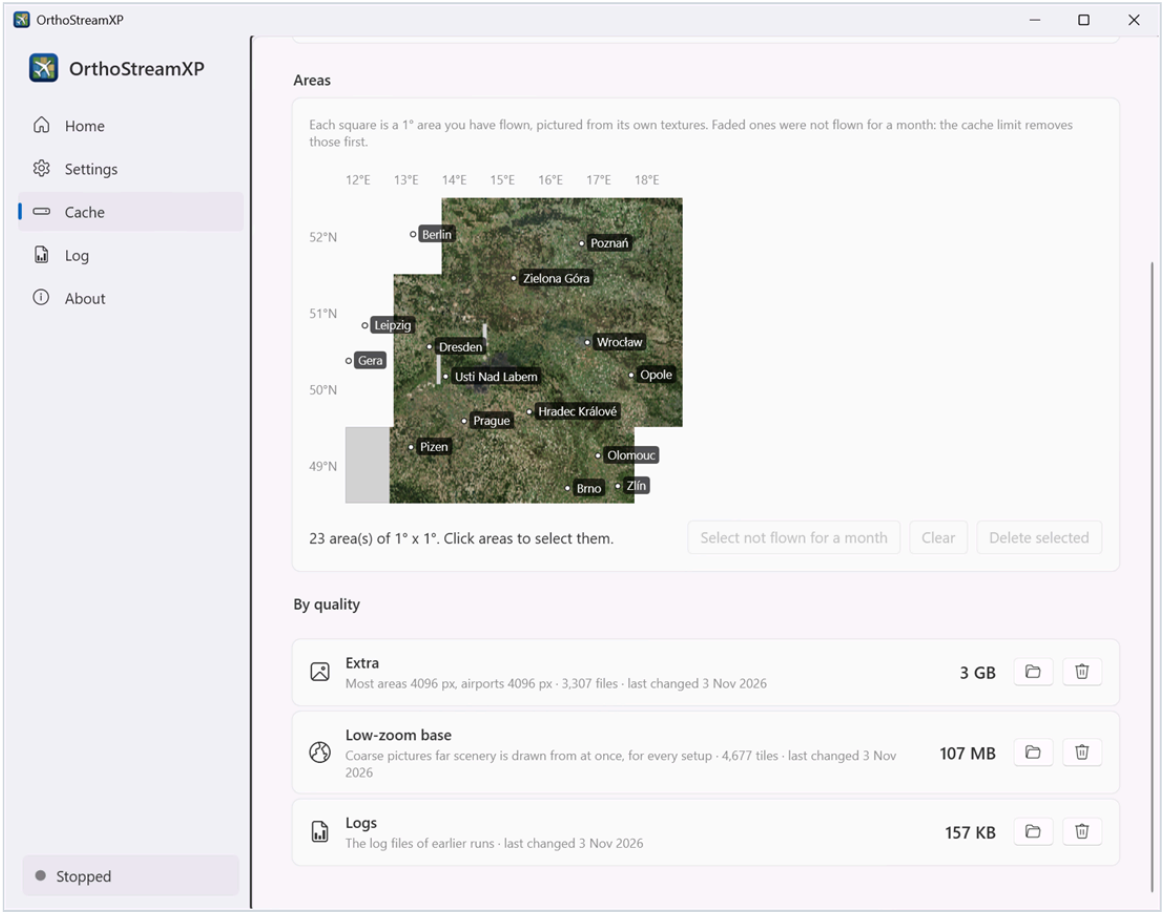
The map shows each 1° × 1° area you have flown, pictured from its own textures. Faded squares were not flown for a month; the cache limit removes those first.

To free space by hand:

1. Click the areas you want to remove, or click **Select not flown for a month**.
2. Click **Delete selected**. OrthoStreamXP shows how much space the selected areas take and asks you to confirm. The textures are deleted in every quality. **Clear** empties the selection.

Areas can be deleted only while OrthoStreamXP is stopped, because X-Plane may be reading them.

## By quality




Cache by quality

Each quality preset keeps its textures in its own folder, so switching presets never throws textures away. This list shows how much each one takes:

- **One row per preset you have flown with** (for example *Extra*), with its texture sizes and number of files. The preset you use now is marked **In use**. Delete a preset you no longer use to free its space.
- **Low-zoom base:** coarse pictures that distant scenery is drawn from. All presets share it. If you delete it, distant scenery is a plain colour at the start of a flight until these pictures are downloaded again.
- **Logs:** the log files of earlier runs.
- A folder left by an older version of OrthoStreamXP, from before each preset had its own folder, is listed too. OrthoStreamXP no longer uses it, so you can delete it.

Each row has a folder button that opens it in Explorer and a bin button that deletes it after you confirm.

Next: Troubleshooting

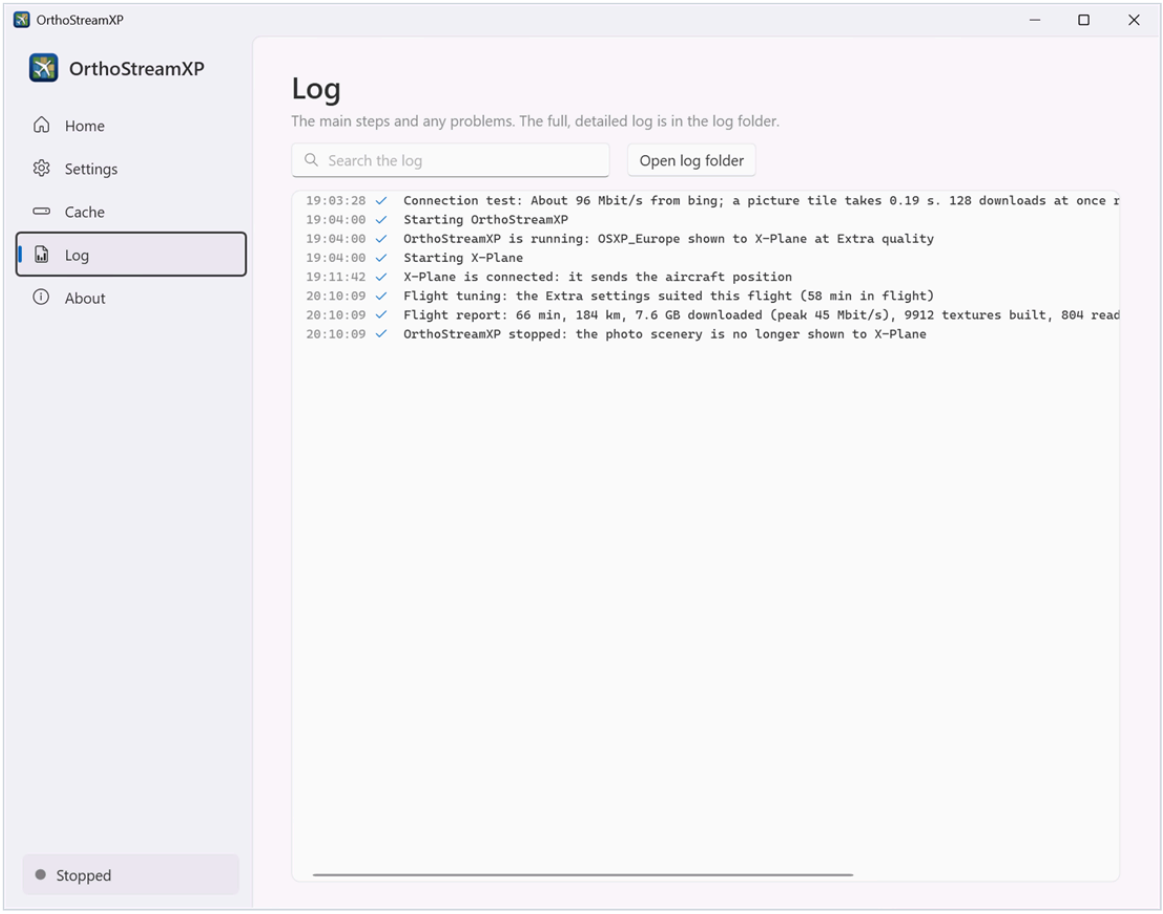
# Troubleshooting

When something goes wrong, Home shows a notice that says what happened, usually with a button for the fix.

## Common problems

| Notice or symptom | What to do |
|---|---|
| **No photos in X-Plane** | Make sure OrthoStreamXP was running **before** X-Plane started. If Home says **Restart X-Plane**, quit X-Plane and start it again. |
| **WinFsp is needed** / **WinFsp could not show the scenery** | Click **Get WinFsp**, install (or repair) WinFsp, restart your computer, then click **Start** again. If you installed WinFsp or OrthoStreamXP and have not restarted Windows since, restart it first. |
| **A scenery folder is in the way** | X-Plane's `Custom Scenery` folder already has a folder with the same name as one of your packages. Usually another streaming tool still shows it there: close that tool and click **Start** again. If it is a real folder, move or rename it. **Open Custom Scenery** opens the folder. |
| **Your scenery packages were not found** | The packages folder is missing or empty. Click **Open settings** and choose the parent folder that holds your packages (each with an `Earth nav data` folder). |
| **X-Plane was not found** | The X-Plane folder has no `Custom Scenery` folder. Choose the folder X-Plane is installed in. |
| **A folder cannot be written** | Choose a cache folder you can write to, for example one in your user folder. |
| **A setting needs attention** | A value in the settings file is not valid. The notice names it; click **Open settings** to fix it. |
| **Setup is not finished** | Click **Run setup**. |
| **X-Plane could not be started** | **Start and fly** could not start X-Plane. The photo scenery is ready: start X-Plane yourself. |
| **Restart OrthoStreamXP to use your new settings** | The settings file changed while OrthoStreamXP was running. Click **Restart now** to use the new values. |
| **Downloads are failing** | Check your internet connection. OrthoStreamXP keeps trying; **Open log** shows the details. |
| **Blurry ground for a while** | A new area takes a few minutes the first time. If it happens often, try a lower preset or apply the tuning suggestions after your flight (see Flying). |
| **Patchy colours between photos** | Turn on **Even out the colours between photos** in Settings › Colours. |
| **Photos look too bright next to X-Plane's own scenery** | Lower **Exposure** in Settings › Colours. |

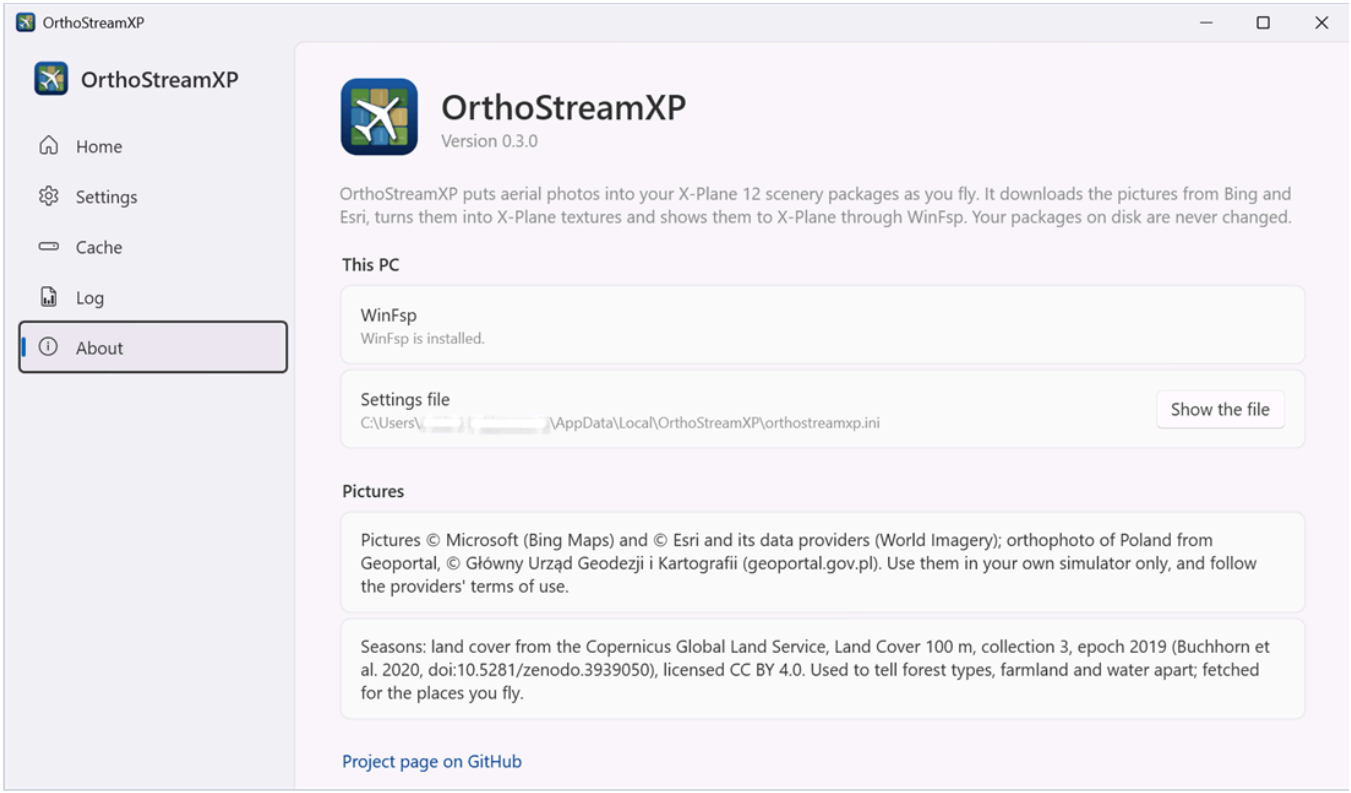
## The Log page




The Log page

The Log page lists the main steps and any problems of this session. Type in **Search the log** to filter the list. **Open log folder** opens the full, detailed logs in `%LOCALAPPDATA%\OrthoStreamXP\logs`.

## The About page




The About page

The About page shows your version, whether WinFsp is installed (with a **Get WinFsp** button if it is not), where the settings file is, the imagery credits, and a link to the project page on GitHub.

## Updates

OrthoStreamXP checks for updates when it opens and from time to time while it runs, and downloads a new version in the background. When the update is ready, Home shows **OrthoStreamXP x.y.z is ready to install**. If the scenery is streaming, finish your flight and stop it first; then click **Update and restart**. Your settings, cache and flight history stay.

## Uninstalling

Use **Windows Settings › Apps**. OrthoStreamXP asks whether to delete your settings and downloaded scenery as well. If you answer **No**, they stay for a later install.

## Reporting a bug

Open an issue on the project's GitHub page and include:

- **What happened:** what you saw, and what you expected instead.
- **Steps to reproduce:** what you did before the problem appeared, for example the area you were flying over and your quality preset.
- **Your system:** OrthoStreamXP version (on the About page), Windows version and graphics card.
- **The session log:** the log file of that session, from `%LOCALAPPDATA%\OrthoStreamXP\logs`.